



Fairness testing for uplift models

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Abstract

Uplift modeling was first initiated in the industry in the early 2000s as a new methodology to improve marketing efficiency by predicting individual treatment effect (ITE). It estimates the conditional average treatment effect (CATE) as the difference in outcome probabilities with and without treatment. Recently, AI ethics including fairness evaluation has received significant attention from academia, industry, and regulatory agencies. However, standard fairness metrics, originally developed for conventional predictive models, generally require ground truth (ITE) and cannot be applied directly for uplift models. In this paper, we propose a novel and practical approach to compute fairness metrics suitable for uplift models. A formal framework is first established based on probability theory. It is followed by a simulation analysis to demonstrate its effectiveness. Finally, we illustrate how to apply the approach through an example using public data.

Keywords Uplift modeling · Individual treatment effect · Conditional average treatment effect · Fairness metrics · Equal opportunity · True positive rate · AI ethics · Responsible AI

Introduction

Corporations have used data and advanced analytics to cultivate customer relationships since early 1980s (Jackson and Wang 1996; Roberts and Berger 1999; Storey and Cohen 2002; Maex and Brown 2012). A key tactic in that effort is to develop response models to predict customers or prospects who are likely to respond to marketing offers. While traditional response models can identify customers who are likely to take desirable actions after receiving a marketing touch, these models cannot distinguish between customers who will not purchase unless they receive marketing offers (those marketers should target) and those

who will purchase regardless of offers (those marketers should not target).

To overcome this limitation, Lo (2002) proposed a **true-lift** modeling approach where the objective is to identify customers whose decisions will be *positively influenced* by marketing efforts. The methodology is also known as **uplift** modeling (Radcliffe and Surry 1999; Radcliffe 2007; Oechsle 2023; Sanisoglu et al. 2023) or **net lift** (Lund 2012; Kubiak 2012). This approach has since been widely used to improve program efficiency because it enables marketers to target customers who will most likely be influenced by marketing campaigns (e.g., email, print, outbound phone channel, digital display, webinars) and has been also applied in political election and medicine (Siegel 2011, 2013; Yong 2015).

With the increasing importance of AI ethics,¹ it is crucial that all AI models are evaluated for fairness before they are deployed for decision making in marketing and sales applications. Such a practice has been evolving from recommendation to requirement (e.g., White House Blueprint for an AI Bill of Rights 2022; Schwartz et al. 2022;

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¹ While fairness is a key component of AI ethics, Responsible AI, or Trustworthy AI and is the focus of this paper, AI ethics includes other areas such as explainability, safety, and privacy (e.g., White House Blueprint for an AI Bill of Rights 2022; Blackman 2022; Khaya et al. 2024).



Table 1 A typical confusion matrix

	Predicted label	
	Yes: $\hat{Y} = 1$	No: $\hat{Y} = 0$
<i>True label</i>		
Yes: $Y = 1$	True positive (TP)	False negative (FN)
No: $Y = 0$	False positive (FP)	True negative (TN)
Total	Predicted positive ($PP = TP + FP$)	Predicted negative ($PN = FN + TN$)

ISO/IEC 2023; Groves et al. 2024; Senate Bill 24-205 2024). While it is straightforward to assess fairness for conventional response models using standard fairness metrics, these metrics typically require ground truth² and cannot be applied to uplift models where the ground truth is not available due to the *Fundamental Problem of Causal Inference* [i.e., an individual can be in either treatment or control but not in both, from Holland (1986), see also Brumback (2022, Chap. 3)]. In this paper, we propose a new methodology to enable fairness testing for uplift models under the assumption that the probability of having a positive lift exceeds that of a negative lift (formally stated in “Existing fairness testing approaches for uplift modeling” section).

Our paper is outlined as follows: we review the relevant literature on fairness metrics, uplift modeling, and their intersection in “Literature review” section, followed by our proposed methodology in “Methodology” section. A simulation analysis in “Empirical evaluation of approaches” section demonstrates the effectiveness of the approach. Additionally, we carry out fairness testing based on the proposed methodology using an actual data set in “A case study” section, followed by the conclusion and discussion in “Conclusion and discussion” section.

Literature review

Fairness metrics

Fairness in AI has been recently posed as a challenge for businesses across many industries. McKinsey points out widespread concern and skepticism on how to imbue AI systems with human ethical judgment given that even seemingly innocuous decisions can have grave implications (Burkhardt et al. 2019). Current research into AI bias suggests that most AI models, if not properly vetted for unintentional bias, will have different outcomes for individuals of different ethnicities, gender, sexual orientation, or age, even if those protected traits have no true impact on the outcome the model is attempting to predict (Mehrabi

et al. 2021). Group Fairness is concerned with achieving parity of positive outcomes or machine learning decisions across groups, see Dwork and Ilvento (2018). In this setting, groups refer to protected and unprotected groups, e.g., different gender or racial groups. For example, we want a classifier to predict a similar proportion of individuals receiving loan approval irrespective of their race attribute, referred to as equality of opportunity in Hardt et al. (2016). Note that optimizing for fairness may lower the accuracy of the AI model, i.e., a trade-off between algorithmic fairness and algorithmic performance may exist (Dwork et al. 2017; Caton and Haas 2024). Most of the existing research in algorithmic fairness focuses on measuring group fairness in binary classification cases, see e.g., Hajian et al. (2011), Hardt et al. (2016), Bonchi et al. (2016), Menon and Williamson (2017), Berk et al. (2018), Buolamwini and Gebu (2018), Breger (2020), Rodolfa et al. (2020), and Lo et al. (2024).

Fairness metrics are derived from several model accuracy measures in a typical confusion matrix (Table 1) where actual and predicted labels are denoted by Y and $\hat{Y}$, respectively. Those measures include: True Positive Rate $TPR = \frac{TP}{TP+FN}$, False Positive Rate (FPR) = $\frac{FP}{FP+TN}$, False Negative Rate (FNR) = $1 - FPR$, number of Predictive Positives $PP = TP + FP$ and number of Predictive Negatives $PN = FN + TN$, where TP , TN , FP , and FN are the numbers of true positives, true negatives, false positives, and false negatives, respectively.

Standard fairness metrics are listed in Table 2 where the subscripts in the definition column indicate the demographic group (0 or 1). More complete surveys of numerous AI fairness metrics with bias mitigation methods can be found in Caton and Haas (2024) and Kheya et al. (2024). In particular, Kheya et al. (2024) notice that the majority of the biases can be mitigated using techniques such as sampling, re-weighting, and causal reasoning.

Uplift modeling

In the early days of database marketing, response models were typically developed to target audience in acquisition or cross-sell campaigns. Response models are created using supervised learning algorithms to predict a target variable

² Commonly defined as actual value of the target variable, see ISO/IEC (2022), for example.



Table 2 Common fairness metrics

Metric name	Definition	Possible range	Ideal value	Conventional fairness range [guided by the fourth-fifth rule from the US EEOC (1978), see also Feldman et al. (2015)]
Statistical parity (<i>SP</i>) or demographic parity	$\frac{PP_1}{PP_1+PN_1} - \frac{PP_0}{PP_0+PN_0}$	$[-1, 1]$	0	$[-0.2, 0.2]$
Disparate impact (<i>DI</i>) or disparate impact ratio	$\frac{PP_1}{PP_1+PN_1} / \frac{PP_0}{PP_0+PN_0}$	$[0, \infty)$	1	$[0.8, 1.2]$ or $[0.8, 1.25]$
Equal opportunity (<i>EO</i>) or recall parity	$TPR_1 - TPR_0$	$[-1, 1]$	0	$[-0.2, 0.2]$
False negative rate (<i>FNR</i>) parity	$FNR_1 - FNR_0 = -EO$	$[-1, 1]$	0	$[-0.2, 0.2]$
Predictive equality (<i>PE</i>) or false positive rate (<i>FPR</i>) parity	$FPR_1 - FPR_0$	$[-1, 1]$	0	$[-0.2, 0.2]$
Average odds (<i>AO</i>) or average odds difference	$\frac{EO+PE}{2}$	$[-1, 1]$	0	$[-0.2, 0.2]$

(responders vs non responders) for a modeling population (customers or prospects in a previous marketing campaign) using transactional and demographic data as features. Response models served the industry well in cases where the baseline take-action rates are relatively low for the untreated group.

As the industry matured, marketers and data scientists realized a flaw in the traditional approach in situations where baseline response rates are nontrivial. In a hypothetical example (Table 3) where the responses rates for treated individuals in model and random groups are 1% and 0.4%, respectively, the model appears to show a significant gain of model performance over random. However, assuming individuals who are not treated (no mail) exhibit the same response rates as their treated counterparts, the incremental values driven by the treatment are zero for both the modeled group and the random group, resulting in a waste of marketing budget.

To overcome this deficiency of traditional response modeling, researchers such as Radcliffe and Surry (1999) and Lo (2002) propose various techniques to find the customers whose decisions will be *positively* influenced by marketing interaction. More formally, uplift modeling refers to a class of predictive modeling approaches where the objective is to predict the conditional average treatment effect (CATE) defined as the difference in the probability of an outcome when a customer receives a treatment over

the baseline probability when he/she does not receive any treatment, given individual attributes. See Lo (2008), Gubela et al. (2017), and Haughton et al. (2025) for further details.

A data mining usage report by Rexer (2012) mentions that uplift/true-lift modeling is more popular among corporate practitioners and consultants than in academic or government agencies. The methodology has been widely leveraged by marketers to improve program efficiency. In the 2012 presidential campaign, uplift modeling was utilized to target campaign donors and ultimately voters, see Scherer (2012), Siegel (2013), and Stedman (2013) which report the significance of the methodology to the election outcome. Similar methodology has been applied in personalized medicine (Yong 2015). See Devriendt et al. (2018) for a survey of uplift modeling techniques.

Existing fairness testing approaches for uplift modeling

The first study to address fairness for uplift modeling is by Michalsky and Kadioglu (2021) but the approach they propose demands (1) additional data from a subsequent experiment; and (2) surrogate ground truth be constructed based on uplift model scores. In practice, a relatively simplistic approach is often adopted where observed take-actions are used to substitute the unknown ground truth.

Table 3 Illustrative example of marketing campaign measurement

	Treatment (e.g., mail)	Control (e.g., no mail) % response rate	Increment (treatment – control)
Model	1.0	1.0	0.0
Random	0.4	0.4	0.0



This approach is defined more formally as the *Naïve* estimator in “[Approximation of the ground truth for assessing model fairness](#)” section and Appendix 6. Its lack of accuracy is illustrated in “[Empirical evaluation of approaches](#)” section.

To overcome the limitations of existing approaches, we propose a new methodology where fairness metrics can be derived as soon as the uplift model is developed without the need for additional data or assumptions. With this new methodology, fairness testing can be conducted for uplift models in a similar way as with conventional response models with a high degree of accuracy.

In subsequent sections, we will derive Equal Opportunity (*EO*) for uplift models. *EO* is conditional on the ground truth and commonly regarded as a key metric. Notice that *FNR* Parity is simply $-EO$, and *PE* will be described in Appendix 4 using a similar approach. *AO* is the simple average of *EO* and *PE*. *SP* and *DI* are straightforward for uplift modeling and will be covered briefly in “[Methodology](#)” section.

Methodology

Unique challenge with measuring fairness for uplift models

Assuming data from a randomized controlled trial (RCT) is available and the sample represents the target population, the Individual Lift or individual treatment effect (ITE) for individual i is defined as:

$$L_i = Y_{1i} - Y_{0i}, \quad (1)$$

where Y_{1i} and Y_{0i} represent the *potential outcomes* for treatment and control, respectively, and assumed to be binary. This convention is standard in the causal inference literature where potential outcomes are represented by a pair of random variables that are specific to treatment or control, as described by Imbens and Rubin (2015, Chap. 1), Morgan and Winship (2015, Chap. 2), or Angrist and Pischke (2009, Chap. 2). This means that if the individual is in the treatment group, the outcome would be Y_{1i} . Otherwise, the outcome would be Y_{0i} , and each individual is in either the treatment or control group but not both. In this convention, the treatment has a positive effect over control when $L_i = 1$ and it has no positive effect when $L_i < 1$.

To interpret the potential outcomes (Y_{1i}, Y_{0i}) , consider an experiment involving identical twins where one is given a lot of cookies daily and the other is not. If cookie intake causes the outcome of weight gain, one expects the ITE to be positive. If the sample consists of many pairs of identical twins, we expect $L_i > 0$ in most cases (assuming no confounders exist for them).

A common interest in measuring the effect from an RCT experiment is the average treatment effect (ATE), which is the mean or expected value of (1) across the population:

$$E(L_i) = E(Y_{1i} - Y_{0i}) = p_1 - p_0, \quad (2)$$

where p_1 and p_0 are treatment response rate and control response rate, respectively. ATE can be easily estimated using RCT data since p_1 and p_0 can be estimated by the sample response rates for the treatment and control groups, respectively.

While estimating the overall ATE is useful in determining the population level treatment effectiveness, marketers are often interested in the ATE for various subgroups, which can be defined as the CATE as described by Abrevaya et al. (2015) and Jacob (2021). More formally, CATE is the *conditional* expectation of ATE:

$$E(L_i|X_i = x) = E(Y_{1i} - Y_{0i}|X_i = x) = p_1(x) - p_0(x), \quad (3)$$

where $p_1(x)$ and $p_0(x)$ are the true treatment and control response probabilities, respectively, as a function of features or covariates, x .

Unlike ATE in (2) which can be directly estimated by sample proportions, CATE is not directly observable and requires estimation through predictive modeling or machine learning. If we can estimate $E(L_i|X_i = x)$, we can use it to prioritize marketing efforts on those who are expected to be influenced most by marketing interventions. In fact, estimating CATE is the essence of uplift modeling. There are multiple modeling techniques available to model CATE as a function of x , see Radcliffe and Surry (1999), Lo (2002), Kane et al. (2014), Athey and Imbens (2015, 2016), and Zhao et al. (2017). The goal of our paper is to address how to assess fairness of uplift models regardless of the modeling techniques.

It is important to note that the ground truth for ITE, L_i , is unknown in reality due to the *Fundamental Problem of Causal Inference*, i.e., the treatment effect for a given individual is not observable because we cannot observe *both* the treatment and control outcomes from the same person, see Holland (1986) and Imbens and Rubin (2015, Chap. 1). Not having the ground truth poses a *unique* challenge for assessing fairness because most fairness metrics require ground truth data as inputs. In the following sections, we propose a novel approach to conduct fairness testing for uplift models.

Approximation of the ground truth for assessing model fairness

We first consider Statistical Parity (*SP*) which is the acceptance or selection rate parity (i.e., the difference in the probability of individuals being classified positively



between two demographic groups) and is one of the simplest metrics:

$$\begin{aligned} SP &= P(\text{Top scores ranked by } S|A = 1) \\ &\quad - P(\text{Top scores ranked by } S|A = 0) \\ &= P(S \geq k|A = 1) - P(S \geq k|A = 0), \end{aligned}$$

where S = model score, and assuming a threshold k is established by marketers for selecting a target audience.

Note that $P(S \geq k|A = a)$ is the proportion of individuals in demographic group $A = a$ whose scores are at least k , which can be computed from the data without the ground truth of individual level potential outcomes. Hence, it is straightforward to obtain SP in an uplift setting. Similarly, it is easy to compute Disparate Impact (DI) which is simply the ratio $\frac{P(S \geq k|A=1)}{P(S \geq k|A=0)}$ as opposed to the difference metric in SP .

It is straightforward to compute Statistical Parity and Disparate Impact in uplift models because they do not require ground truth. However, the same is not true for all other metrics in Table 2. We now focus on a commonly used metric, **Equal Opportunity (EO)**, for which we need to know for whom the treatment is beneficial, i.e., $L_i = 1$ versus not beneficial, i.e., $L_i < 1$.³ Assume a model score, S_i , is available at the individual level through uplift modeling which estimates CATE in (3) as a function of features x [techniques today include the two-model approach, treatment dummy approach, four-quadrant, uplift tree, uplift random forest, and the meta learners in Uber (2022)]:

$$\begin{aligned} S_i(x) &= \hat{P}(Y_i = 1|T_i = 1, X_i = x) \\ &\quad - \hat{P}(Y_i = 1|T_i = 0, X_i = x) \\ &= \hat{p}_1(x) - \hat{p}_0(x), \end{aligned} \quad (4)$$

where $\hat{\cdot}$ denotes an estimator, Y_i is the observed outcome for i , and $T_i = 1$ or 0 depending on whether the individual

To understand under which cases ground truth is unavailable, Table 4 lists all possible cases and corresponding potential values for the ground truth, L_i . In Case 1, there is no response if treated ($Y_i = Y_{1i} = 0$), the treatment is simply not effective, $L_i < 1$ will always hold regardless of the value of potential outcome Y_{0i} if not treated (control). Likewise, in Case 4, the individuals respond positively without treatment so having treatment cannot make it better, and $L_i < 1$ will always hold. As a result, both Cases 1 and 4 are undesirable from the perspective of looking for a positive effect. In Cases 2 and 3, on the other hand, L_i can either be 1 or < 1 , and the ground truth is unknown. Individuals in these cases are potentially desirable candidates for treatments depending on the values of Y_{0i} and Y_{1i} .

The fundamental challenge of fairness assessment for uplift modeling is the fact that ground truth data is only available for some cases and unavailable for others.

We now consider our proposed approximation for Equal Opportunity (EO) which is the difference in True Positive Rate (TPR) between two groups, see Bellamy et al. (2018) and Barocas et al. (2023):

$$EO = TPR(A = 1) - TPR(A = 0), \quad (5)$$

where A is an attribute such as gender or race, and

$$TPR(A = a) = P(\text{selected targets} | \text{actual good targets}, A = a).$$

The conventional acceptance range for EO is typically $[-0.2, 0.2]$ to be considered fair, guided by the fourth-fifth rule from the US EECO (1978). Assuming a threshold k is established by marketers for selecting a target audience (model score $S \geq k$), then (suppressing subscript i for brevity):

$$\begin{aligned} TPR(A = a) &= P(\text{Top scores ranked by } S|L = 1, A = a) \\ &= P(S \geq k|L = 1, A = a) \\ &= \frac{P(L = 1|S \geq k, A = a)P(S \geq k|A = a)}{P(L = 1|S \geq k, A = a)P(S \geq k|A = a) + P(L = 1|S < k, A = a)P(S < k|A = a)}, \end{aligned} \quad (6)$$

is in the treatment group or not in a randomized experiment.

³ Since marketers are typically interested in which customers are positively influenced by the treatment, we are assuming there is no interest in distinguishing between

$L_i = 0$ (no effect) and -1 (negative effect).

Table 4 All possible observations of (T_i, Y_{1i}, Y_{0i}) and the corresponding observable L_i

Case	T_i	Y_{1i}	Y_{0i}	L_i
1	1	0	NA (0 or 1)	< 1
2	1	1	NA (0 or 1)	$1 \text{ or } < 1$
3	0	NA (0 or 1)	0	$1 \text{ or } < 1$
4	0	NA (0 or 1)	1	< 1



according to Bayes' theorem.

$TPR(A = a)$ in Eq. (6) cannot be directly computed because, while $P(S \geq k | A = a)$ can be obtained from the data, the probabilities of having a positive lift in both regions (above and below the threshold k), $P(L = 1 | S \geq k, A = a)$ and $P(L = 1 | S < k, A = a)$, are unknown (as L_i 's are unobservable).

Without the new approach proposed in this paper, the most intuitive way to compute EO for an uplift model is to use the observed values of Y to represent the unobservable ITEs (as if it is a conventional classification model) along with the uplift model scores to compute $TPRs$. The authors have seen this method adopted by uplift modeling practitioners. We will call this approach the *Naïve estimator*. We argue that this estimator is systematically biased because the observed values of Y do not always accurately represent ITEs in the TPR calculation (see Appendix 6 for a further description of the Naïve estimator).

We proceed with our proposed approach. For simplicity of notations, let H_{1a} be the set of $\{S \geq k, A = a\}$ and H_{2a} be the set of $\{S < k, A = a\}$, which are the individuals in the demographic group $A = a$ whose scores are above (or equal to) or below the threshold, respectively. In other words, H_{1a} and H_{2a} represent the targeted and non-targeted groups, respectively, for $A = a$.

We first establish Lemma 1 which will support the true positive rate approximation in Theorem 1.

Lemma 1 *The probability of having a positive lift is the difference in treatment and control response rates plus a non-negative value (in both the targeted and non-targeted groups).*

Expressed formally, the probability of having a positive effect⁴ in the set of H_{ja} , denoted by

$$\begin{aligned} \lambda_{ja} &= P(L = 1 | H_{ja}) = P(Y_1 = 1, Y_0 = 0 | H_{ja}) \\ &= P(Y_1 = 1 | H_{ja}) - P(Y_0 = 1 | H_{ja}) + \delta_{ja}, \end{aligned} \quad (7)$$

$$\lambda_{ja} \geq P(Y_1 = 1 | H_{ja}) - P(Y_0 = 1 | H_{ja}), \quad (8)$$

where the probability of having a negative effect, $\delta_{ja} = P(Y_1 = 0, Y_0 = 1 | H_{ja}) = P(L = -1 | H_{ja}) \geq 0$.

Additionally, directly from (7),

$$\lambda_{ja} > \delta_{ja}, \quad (9)$$

if and only if

$$\begin{aligned} E(L | H_{ja}) &= P(Y_1 = 1 | H_{ja}) - P(Y_0 = 1 | H_{ja}) > 0, \\ j &= 1, 2, a = 0, 1. \end{aligned} \quad (10)$$

Proof appears in Appendix 1.

⁴ This is also known as the probability of necessity and sufficiency (PNS) in Pearl (1999) and Mueller et al. (2022).

Table 5 Contingency table of potential outcomes

	$Y_0 = 0$	$Y_0 = 1$
$Y_1 = 0$	A	B
$Y_1 = 1$	C	D

The interpretation of (7) is that the probability of having a positive lift is the difference in treatment and control response rates plus a δ_{ja} (a non-negative value) over subset H_{ja} . The treatment and control response rates can be estimated directly from data.⁵ δ_{ja} , on the other hand, cannot be estimated in reality. Note that the value δ_{ja} represents the probability of having a negative lift, a value that intuitively should be relatively small especially in the marketing and sales context.⁶

To further understand δ_{ja} , we consider the joint probability distribution of (Y_1, Y_0) in subset H_{ja} in Table 5, where δ_{ja} is represented by Quadrant B in the table. If treatment has an overall positive effect (i.e., $E(Y_1 - Y_0 | H_{ja}) = P(Y_1 = 1 | H_{ja}) - P(Y_0 = 1 | H_{ja}) > 0$), from (7), the probability of having a positive lift (Quadrant C), λ_{ja} , will be greater than the probability of having a negative lift (Quadrant B), δ_{ja} . For an intuitive explanation, imagine a sibling comparison experiment⁷ using identical twins is conducted to measure the effects of consuming cookies on body weights. The experiment is set up such that one of the siblings among pairs of identical twins (treatment) eats cookies daily and the other does not eat any (control). Let Y_1 and Y_0 represent their resulting weights at the end of the trial for treatment and control, respectively. If we expect cookie intake to have a positive impact on weight gain, Quadrant B's associated probability would be small relative to Quadrant C's (unless eating cookies reduces weight). In other words, we expect

⁵ Applying the exchangeability or ignorability property of randomized experiments (i.e., $Y_0, Y_1 \perp T$) along with the standard consistency theorem or assumption which states that an individual's potential outcome under their observed exposure is exactly the same as their observed outcome, we have $P(Y_1 = 1 | H_{ja}) = P(Y_1 = 1 | T = 1, H_{ja}) = P(Y = 1 | T = 1, H_{ja})$ and $P(Y_0 = 1 | H_{ja}) = P(Y_0 = 1 | T = 0, H_{ja}) = P(Y = 1 | T = 0, H_{ja})$ which are both estimable from data (typically from test data as in fairness evaluation for standard predictive models), see Cole and Frangakis (2009), Pearl (2010), and Hernan and Robins (2024, Chap. 3).

⁶ Search string "negative treatment effects of marketing" returned no publications in Google. The only paper, Simonson, Carmon, and O'Curry (1994), is a special case where promoting a new product feature may decrease likelihood to purchase for certain segments of consumers.

⁷ A design that can control for genetic and shared environmental confounding factors (Saunders et al. 2019).



$\lambda_{ja} > \delta_{ja}$,⁸ as in (9), a condition that directly follows from (7) as long as $P(Y_1 = 1|H_{ja}) - P(Y_0 = 1|H_{ja}) > 0$ as in (10), which is a key requirement for our proposed approximation method in Theorem 1.

Assumption 1 The group level lift is assumed to be positive in both targeted and non-targeted groups, i.e.,

$$P(Y_1 = 1|H_{ja}) - P(Y_0 = 1|H_{ja}) > 0, \quad j = 1, 2, a = 0, 1, \quad (11)$$

where H_{1a} and H_{2a} represent the targeted and non-targeted groups, respectively.

Or equivalently, from Lemma 1, the probability of having a positive lift is assumed to be greater than the probability of having a negative lift in both targeted and non-targeted groups, i.e., $\lambda_{ja} > \delta_{ja}$ (9).

Applying (7) from Lemma 1 to (6), we have

TPR in (12) can be approximated as follows:

$$\widehat{TPR}(A = a) = \frac{1}{1 + \frac{Q_{2a}}{Q_{1a}} O_{Sa}}, \quad (13)$$

where all values in (13) can be estimable from data, if any of the following conditions are satisfied:

- (I) The probabilities of having a negative lift, δ_{1a} and $\delta_{2a} \approx 0$;
- (II) The two ratios of positive lift probability to negative lift probability are close to each other, i.e., $\lambda_{1a}/\delta_{1a} \approx \lambda_{2a}/\delta_{2a}$, or
- (III) The ratios of positive lift probability to negative lift probability, λ_{1a}/δ_{1a} and λ_{2a}/δ_{2a} are both large.

$$\begin{aligned} TPR(A=a) &= P(\text{Top scores ranked by } S|L=1, A=a) = P(S \geq k|L=1, A=a) \\ &= \frac{[P(Y_1 = 1|H_{1a}) - P(Y_0 = 1|H_{1a}) + \delta_{1a}]P(S \geq k|A=a)}{[P(Y_1 = 1|H_{1a}) - P(Y_0 = 1|H_{1a}) + \delta_{1a}]P(S \geq k|A=a) + [P(Y_1 = 1|H_{2a}) - P(Y_0 = 1|H_{2a}) + \delta_{2a}]P(S < k|A=a)} \\ &= \frac{1}{1 + \frac{Q_{2a} + \delta_{2a}}{Q_{1a} + \delta_{1a}} O_{Sa}}, \end{aligned} \quad (12)$$

where $Q_{2a} = P(Y_1 = 1|H_{2a}) - P(Y_0 = 1|H_{2a})$, $Q_{1a} = P(Y_1 = 1|H_{1a}) - P(Y_0 = 1|H_{1a})$ and $O_{Sa} = \frac{1 - P(S \geq k|A=a)}{P(S \geq k|A=a)}$ which is the ratio of model-driven non-target proportion to target proportion for a given demographic group.

Notice that Q_{2a} and Q_{1a} are the differences in treatment and control response rates in the two regions (with scores above and below the threshold k). It is expected that $Q_{1a} > Q_{2a} > 0$, where the first inequality follows from an uplift model to perform reasonably (i.e., selected targets have a higher lift than others) and the second inequality follows from Assumption 1. While Q_{2a} and Q_{1a} are estimable from data (see footnote 5), the values of δ_{1a} and δ_{2a} in Eq. (12) are still unknown and not estimable. We now offer an approximation formula for (12).

Theorem 1 Approximation Formula for TPR.

Our proposed approximation for TPR in uplift models is based on the exact equation from (12) with the removal of the unknown and non-estimable values (δ_{1a} and δ_{2a}). Mathematically,

Proof From Eq. (12), condition I is obvious. For the other conditions, we can compare the relative difference between Eqs. (12) and (13):

$$\begin{aligned} \frac{TPR(A=a) - \widehat{TPR}(A=a)}{TPR(A=a)} &= \frac{\frac{Q_{2a}}{Q_{1a}} O_{Sa} - \frac{Q_{2a} + \delta_{2a}}{Q_{1a} + \delta_{1a}} O_{Sa}}{1 + \frac{Q_{2a}}{Q_{1a}} O_{Sa}} \\ &= \frac{1}{(1 + \frac{Q_{2a}}{Q_{1a}} O_{Sa})} \frac{Q_{2a}}{Q_{1a}} O_{Sa} \left(1 - \frac{1 + \delta_{2a}/Q_{2a}}{1 + \delta_{1a}/Q_{1a}} \right) \\ &= \kappa_a \left(\frac{\frac{1}{\delta_{1a}} - 1}{1 + \frac{1}{\delta_{1a}}} - \frac{\frac{1}{\delta_{2a}} - 1}{1 + \frac{1}{\delta_{2a}}} \right) \end{aligned}$$

by Eq. (7) in Lemma 1: $\lambda_{ja} = Q_{ja} + \delta_{ja}$, $j = 1, 2$, $a = 0, 1$; where $\kappa_a = \frac{1}{(1 + \frac{Q_{2a}}{Q_{1a}} O_{Sa})} \frac{Q_{2a}}{Q_{1a}} O_{Sa}$. From Assumption 1, $Q_{ja} > 0$, or equivalently, $\lambda_{ja} > \delta_{ja}$. Conditions II and III then follow.

In the above theorem, Condition I is a stronger one while Condition III is weaker.

In practice for (13), the denominator of O_{Sa} could be very small, or the estimates of Q_{1a} or Q_{2a} may become negative numerically, so a practical computational formula for the true positive rate is

⁸ Note that a lower probability of having a negative lift (δ_{ja}) is a much weaker condition than the monotone treatment response assumption in Manski (1997), where $Y_1 \geq Y_0$ for every individual [see also Flores and Chen (2018, Chap. 3) and Neal (2020, Chap. 8)].



$$\widehat{TPR}(A = a) = \frac{\max\{0, Q_{1a}\}P(S \geq k|A = a)}{\max\{0, Q_{1a}\}P(S \geq k|A = a) + \max\{0, Q_{2a}\}(1 - P(S \geq k|A = a))}. \quad (14)$$

In addition to a TPR estimate in (14),⁹ Theorem 2 below offers a bound for TPR and also a bound for the relative error for the approximate TPR from Theorem 1, which enables us to assess how accurate the approximation would be in practice. We will first need Lemma 2 to support the proof of Theorem 2.

Lemma 2 *Bound for ratio of positive lift probability to negative lift probability.*

$$\frac{\lambda_{ja}}{\delta_{ja}} = \frac{P(Y_1 = 1, Y_0 = 0|H_{ja})}{P(Y_1 = 0, Y_0 = 1|H_{ja})} \geq \text{Max}\left(\frac{p_{tja}}{1 - p_{tja}}, \frac{1 - p_{cja}}{p_{cja}}\right), \quad (15)$$

$$Q_{2a} = P(Y_1 = 1|H_{2a}) - P(Y_0 = 1|H_{2a}),$$

$$Q_{1a} = P(Y_1 = 1|H_{1a}) - P(Y_0 = 1|H_{1a}),$$

$$O_{Sa} = \frac{1 - P(S \geq k|A = a)}{P(S \geq k|A = a)}, \text{ and}$$

$$\kappa_a = \frac{1}{(1 + \frac{Q_{2a}}{Q_{1a}} O_{Sa})} \frac{Q_{2a}}{Q_{1a}} O_{Sa} (< 1),$$

$$M_{ja} = \text{Max}\left(\frac{p_{tja}}{1 - p_{tja}}, \frac{1 - p_{cja}}{p_{cja}}\right), \quad j = 1, 2. \quad (18)$$

Proof is in Appendix 3.

Similar to (14), a more practical computational formula for the bound in (18) is¹⁰:

$$\begin{aligned} & \frac{\max\{0, Q_{1a}\}P(S \geq k|A = a)}{\max\{0, Q_{1a}\}P(S \geq k|A = a) + \max\{0, Q_{2a}\}\left(1 + \frac{1}{M_{2a}-1}\right)(1 - P(S \geq k|A = a))} \\ & \leq TPR(A = a) \leq \frac{\max\{0, Q_{1a}\}P(S \geq k|A = a)}{\max\{0, Q_{1a}\}P(S \geq k|A = a) + \max\{0, Q_{2a}\}\max\left(0, 1 - \frac{1}{M_{1a}}\right)(1 - P(S \geq k|A = a))} \end{aligned} \quad (19)$$

where $p_{tja} = P(Y_1 = 1|H_{ja})$ and $p_{cja} = P(Y_0 = 1|H_{ja})$, $j = 1, 2$, which can be estimated by $P(Y = 1|T = 1, H_{ja})$ and $P(Y = 1|T = 0, H_{ja})$, respectively, using RCT data.

Proof is in Appendix 2 with bound assumptions stated.

Theorem 2 *Bound for the actual TPR*

$$\frac{1}{1 + \frac{Q_{2a}}{Q_{1a}}\left(1 + \frac{1}{M_{2a}-1}\right)O_{Sa}} \leq TPR(A = a) \leq \frac{1}{1 + \frac{Q_{2a}}{Q_{1a}}\left(1 - \frac{1}{M_{1a}}\right)O_{Sa}}. \quad (16)$$

Or, equivalently, the relative error bound for the approximate TPR is:

$$-\frac{\kappa_a}{M_{2a} - 1} \leq \frac{TPR(A = a) - \widehat{TPR}(A = a)}{TPR(A = a)} \leq \frac{\kappa_a}{M_{1a}} \quad (17)$$

where

⁹ In the unusual scenarios where the estimates of both Q_{1a} and Q_{2a} are non-positive, it means the model does not have positive lift for group a in the targeted and non-targeted groups, and in which case $\widehat{TPR}(A = a)$ can be assigned to 0.

To illustrate (14) and (19) with a numerical example for a given demographic group $A = a$, and assuming the top 3 deciles are the selected targets for $A = a$, i.e., $P(S \geq k|A = a) = 0.3$.¹¹:

In $H_{1a} = \{S \geq k, A = a\}$, $p_{t1a} = P(Y_1 = 1|H_{1a})$ is estimated by $P(Y = 1|T = 1, H_{1a}) = 0.35$ (assumed given) and $p_{c1a} = P(Y_0 = 1|H_{1a})$ is estimated by $P(Y = 1|T = 0, H_{1a}) = 0.20$ (assumed given).

Hence, $Q_{1a} = 0.35 - 0.20 = 0.15$.

From (18), $M_{1a} = \text{Max}\left(\frac{0.35}{1-0.35}, \frac{1-0.20}{0.20}\right) = 4$.

¹⁰ Note that in many marketing and sales applications, the response probabilities do not exceed 0.5 whether they are treated or not, i.e., $p_{tja} \leq 0.5$ and $p_{cja} \leq 0.5$, $j = 1, 2$, resulting in $\frac{p_{tja}}{1-p_{tja}} \leq \frac{1-p_{cja}}{p_{cja}}$, and hence, (18) could be further simplified as:

$$M_{ja} = \text{Max}\left(\frac{p_{tja}}{1-p_{tja}}, \frac{1-p_{cja}}{p_{cja}}\right) = \frac{1-p_{cja}}{p_{cja}}.$$

¹¹ Numbers in *italic* in this example are assumed to be available from data. All other numbers are computed.



Similarly, in $H_{2a} = \{S < k, A = a\}$, $p_{t2a} = P(Y_1 = 1|H_{2a})$ is estimated by $P(Y = 1|T = 1, H_{2a}) = 0.15$ (assumed given) and $p_{c2a} = P(Y_0 = 1|H_{2a})$ is estimated by $P(Y = 1|T = 0, H_{2a}) = 0.10$ (assumed given).

Hence, $Q_{2a} = 0.15 - 0.10 = 0.05$.

From (18), $M_{2a} = \text{Max}(\frac{0.15}{1-0.15}, \frac{1-0.10}{0.10}) = 9$.

Using the approximation in (14), we have $\widehat{TPR}(A = a) = \frac{0.15 \times 0.3}{0.15 \times 0.3 + 0.05 \times 0.7} = 0.563$.

The intermediate computations and the output values including bounds of $\widehat{TPR}(A = a)$ using (19) and bounds of the relative error using (17) are summarized in Example 1 in Table 6.

bound, and the relative error is also higher as summarized in Table 6.

More generally, the relative error will be in the range of $[-0.219, 0.146]$ if p_{cja} is less than 0.25 for $j = 1, 2$ and κ_a is 0.438 as above. It can be shown that the bound in (17) for the relative error will be narrower if either response probabilities or $\frac{Q_{2a}}{Q_{1a}}$ (inverse of uplift model performance) is lower.¹²

Finally, applying Theorem 1 to Eq. (5) using (14), Equal Opportunity (EO) can be approximated as follows:

$$\begin{aligned} \widehat{EO} &= \widehat{TPR}(A = 1) - \widehat{TPR}(A = 0) \\ &= \frac{\max\{0, Q_{11}\}P(S \geq k|A = 1)}{\max\{0, Q_{11}\}P(S \geq k|A = 1) + \max\{0, Q_{21}\}(1 - P(S \geq k|A = 1))} \\ &\quad - \frac{\max\{0, Q_{10}\}P(S \geq k|A = 0)}{\max\{0, Q_{10}\}P(S \geq k|A = 0) + \max\{0, Q_{20}\}(1 - P(S \geq k|A = 0))}. \end{aligned} \quad (20)$$

In Example 2 in Table 6, the response probabilities are ten times lower than those in Example 1 (which can be realistic especially in direct marketing), i.e., $(p_{t1a}, p_{c1a}, p_{t2a}, p_{c2a}) = (0.035, 0.020, 0.015, 0.010)$, the $\widehat{TPR}(A = a)$ remains at 0.563 with a much tighter bound

From (19), we can arrive at the following bound for Equal Opportunity (EO) which guarantees its upper and lower limits:

$$\begin{aligned} &\frac{\max\{0, Q_{11}\}P(S \geq k|A = 1)}{\max\{0, Q_{11}\}P(S \geq k|A = 1) + \max\{0, Q_{21}\}\left(1 + \frac{1}{M_{21}-1}\right)(1 - P(S \geq k|A = 1))} \\ &\quad - \frac{\max\{0, Q_{10}\}P(S \geq k|A = 0)}{\max\{0, Q_{10}\}P(S \geq k|A = 0) + \max\{0, Q_{20}\}\max\left(0, 1 - \frac{1}{M_{10}}\right)(1 - P(S \geq k|A = 0))} \\ \leq EO \leq &\frac{\max\{0, Q_{11}\}P(S \geq k|A = 1)}{\max\{0, Q_{11}\}P(S \geq k|A = 1) + \max\{0, Q_{21}\}\max\left(0, 1 - \frac{1}{M_{11}}\right)(1 - P(S \geq k|A = 1))} \\ &\quad - \frac{\max\{0, Q_{10}\}P(S \geq k|A = 0)}{\max\{0, Q_{10}\}P(S \geq k|A = 0) + \max\{0, Q_{20}\}\left(1 + \frac{1}{M_{20}-1}\right)(1 - P(S \geq k|A = 0))}. \end{aligned} \quad (21)$$

(i.e., narrower bound width), and equivalently, the relative error is much lower, see Table 6.

On the other hand, in Example 3 in Table 6, the response probabilities are 30% higher than those in Example 1, $\widehat{TPR}(A = a)$ stays the same but with a wider

¹² This results from the fact that the partial derivative of the upper bound with respect to p_{c1a} or $\frac{Q_{2a}}{Q_{1a}}$ is positive and that of the lower bound with respect to p_{c2a} or $\frac{Q_{2a}}{Q_{1a}}$ is negative.



Table 6 Illustration for Applying the Approximation Formulas

Example	Input values			Intermediate Computations			Output: estimate and error bounds									
	P_{r1a}	P_{c1a}	P_{r2a}	P_{c2a}	Q_{1a}	Q_{2a}	M_{1a}	M_{2a}	κ_a	$\widehat{TPR}$	Lower bound of TPR	Upper bound of TPR	Bound width of TPR	Lower bound of relative error	Upper bound of relative error	Bound width of relative error
1	0.35	0.2	0.15	0.1	0.15	0.05	4,000	9,000	0.438	0.563	0.533	0.632	0.098	− 0.055	0.109	0.164
2	0.035	0.02	0.015	0.01	0.015	0.005	49,000	99,000	0.438	0.563	0.560	0.568	0.008	− 0.004	0.009	0.013
3	0.455	0.26	0.195	0.13	0.195	0.065	2,846	6,692	0.438	0.563	0.522	0.665	0.142	− 0.077	0.154	0.231

Proof Applying the left and right limits from (19) for $TPR(A = 1)$ and $TPR(A = 0)$, we can immediately obtain (21).

Similar approximation formulas are derived for the False Positive Rate (FPR) and Predictive Equality, which is the FPR Parity, see Appendix 4 for the formulas.

Empirical evaluation of approaches

To evaluate the accuracy of the new approach empirically, we generate a dataset simulating a charity donation campaign to 300,000 potential donors, 2/3 in treatment and 1/3 in control, with the following features: gender (a sensitive variable for the fairness, assumed binary), x (representing individual characteristics and following a standard normal distribution). To generate the ground truth, we assume the following flexible model that underlies the utility function:

$$U_t = -1 + \mu t + 4x + \theta \text{ gender} + \phi t \text{ gender}, \quad (22)$$

where t denotes the treatment (1) or control (0) group, μ captures the size of the treatment effect, θ is the gender effect, and ϕ represents the interaction effect between treatment and gender.

The ground truth (unknown in reality) is then obtained through the following standard logit function¹³:

$$P(Y_t = 1) = \frac{\exp(U_t)}{1 + \exp(U_t)} \quad (23)$$

and

$$Y_t \sim \text{Bernoulli}(P(Y_t = 1)) \quad (24)$$

Sixty simulated datasets ($3 \times 4 \times 5 = 60$) are generated for $\theta = 0.5, 1$, and 2 , $\phi = 0, 0.2, 0.6$, and 1 and $\mu = 0.25, 0.5, 1, 2$, and 3 . Each dataset is randomly split into 50% for training the model and 50% as test data to report model performance and compute fairness metrics. More details of our simulation are available in Appendix 5.

An uplift model using the two-model approach¹⁴ is estimated for each simulated dataset using the training data and uplift model scores are computed as the difference in predicted probability between the treatment and control models in the test data. Using the top 3 deciles (top 30%) to establish the target audience, the actual TPR and approximate TPR (i.e., $\widehat{TPR}$) using our proposed method in (14) as well as the actual EO , approximate EO (i.e., $\widehat{EO}$) and its bounds using (20) and (21) are computed for all the simulated datasets using the test data. The results of the Naïve

¹³ Equation 23 transforms raw donation tendency or utility function (U_t) into a probability. Equation 24 transforms the probability into the ground truth (0 or 1).

¹⁴ A full logistic regression without feature selection is used here.



Table 7 Actual, approximate, and Naïve estimator of true positive rates and equal opportunity values with bounds, by main effects of gender and treatment and their interaction effect (actual values are in bold and *italic*)

Simulated dataset	Input parameters			True positive rates			
	1 Gender parameter θ	2 Gender-treatment interaction effect ϕ	3 Treatment effect parameter μ	4 Actual $TPR(A = 1)$	5 Naïve est. of $TPR(A = 1)$	6 Approx $TPR(A = 1)$	7 TPR lower bound ($A = 1$)
1	0.5	0	0.25	<i>0.751</i>	0.417	1.000	1.000
2	0.5	0	0.5	<i>0.703</i>	0.365	0.788	0.495
3	0.5	0	1	<i>0.691</i>	0.317	0.715	0.226
4	0.5	0	2	<i>0.662</i>	0.289	0.643	0.378
5	0.5	0	3	<i>0.627</i>	0.279	0.619	0.467
6	0.5	0.2	0.25	<i>0.852</i>	0.468	1.000	1.000
7	0.5	0.2	0.5	<i>0.809</i>	0.430	0.864	0.592
8	0.5	0.2	1	<i>0.724</i>	0.336	0.707	0.191
9	0.5	0.2	2	<i>0.685</i>	0.305	0.667	0.431
10	0.5	0.2	3	<i>0.634</i>	0.284	0.625	0.491
11	0.5	0.6	0.25	<i>0.922</i>	0.544	0.977	0.849
12	0.5	0.6	0.5	<i>0.903</i>	0.500	0.934	0.601
13	0.5	0.6	1	<i>0.774</i>	0.370	0.751	0.380
14	0.5	0.6	2	<i>0.712</i>	0.327	0.705	0.522
15	0.5	0.6	3	<i>0.660</i>	0.299	0.653	0.542
16	0.5	1	0.25	<i>0.952</i>	0.586	0.980	0.469
17	0.5	1	0.5	<i>0.929</i>	0.536	0.958	0.609
18	0.5	1	1	<i>0.812</i>	0.392	0.796	0.558
19	0.5	1	2	<i>0.732</i>	0.343	0.725	0.589
20	0.5	1	3	<i>0.689</i>	0.313	0.682	0.595
21	1	0	0.25	<i>0.708</i>	0.414	1.000	1.000
22	1	0	0.5	<i>0.731</i>	0.384	0.679	0.167
23	1	0	1	<i>0.676</i>	0.298	0.665	0.320
24	1	0	2	<i>0.660</i>	0.278	0.660	0.484
25	1	0	3	<i>0.626</i>	0.254	0.623	0.516
26	1	0.2	0.25	<i>0.838</i>	0.468	0.799	0.262
27	1	0.2	0.5	<i>0.813</i>	0.429	0.760	0.114
28	1	0.2	1	<i>0.724</i>	0.319	0.696	0.403
29	1	0.2	2	<i>0.674</i>	0.285	0.679	0.520
30	1	0.2	3	<i>0.639</i>	0.261	0.633	0.539
31	1	0.6	0.25	<i>0.916</i>	0.521	0.949	0.698
32	1	0.6	0.5	<i>0.880</i>	0.475	0.868	0.545
33	1	0.6	1	<i>0.777</i>	0.347	0.770	0.576
34	1	0.6	2	<i>0.706</i>	0.305	0.705	0.588
35	1	0.6	3	<i>0.676</i>	0.276	0.676	0.601
36	1	1	0.25	<i>0.946</i>	0.563	0.978	0.927
37	1	1	0.5	<i>0.915</i>	0.512	0.923	0.804
38	1	1	1	<i>0.809</i>	0.375	0.804	0.667
39	1	1	2	<i>0.732</i>	0.319	0.731	0.639
40	1	1	3	<i>0.697</i>	0.288	0.702	0.640
41	2	0	0.25	<i>0.787</i>	0.392	0.624	0.345
42	2	0	0.5	<i>0.774</i>	0.355	0.630	0.397
43	2	0	1	<i>0.701</i>	0.271	0.692	0.553

Table 7 (continued)

Simulated dataset	Input parameters			True positive rates			
	1 Gender parameter θ	2 Gender-treatment interaction effect ϕ	3 Treatment effect parameter μ	4 Actual $TPR(A = 1)$	5 Naïve est. of $TPR(A = 1)$	6 Approx $TPR(A = 1)$	7 TPR lower bound ($A = 1$)
44	2	0	2	0.666	0.241	0.645	0.560
45	2	0	3	0.659	0.226	0.650	0.594
46	2	0.2	0.25	0.867	0.425	0.769	0.588
47	2	0.2	0.5	0.850	0.387	0.810	0.672
48	2	0.2	1	0.729	0.285	0.706	0.583
49	2	0.2	2	0.690	0.252	0.673	0.598
50	2	0.2	3	0.673	0.233	0.657	0.606
51	2	0.6	0.25	0.928	0.478	0.904	0.840
52	2	0.6	0.5	0.889	0.434	0.861	0.779
53	2	0.6	1	0.778	0.315	0.757	0.674
54	2	0.6	2	0.727	0.267	0.719	0.662
55	2	0.6	3	0.694	0.242	0.672	0.632
56	2	1	0.25	0.941	0.514	0.915	0.872
57	2	1	0.5	0.916	0.465	0.899	0.851
58	2	1	1	0.810	0.333	0.803	0.743
59	2	1	2	0.763	0.282	0.754	0.709
60	2	1	3	0.729	0.254	0.705	0.675
Median absolute error					0.407	0.017	
Median absolute percent error					54.63	2.24	

Simulated dataset	True positive rates					
	8 TPR upper bound ($A = 1$)	9 Actual $TPR(A = 0)$	10 Naïve est of $TPR(A = 0)$	11 Approx $TPR(A = 0)$	12 TPR lower bound ($A = 0$)	13 TPR upper bound ($A = 0$)
1	1.000	0.596	0.379	0.598	0.409	0.779
2	0.941	0.656	0.376	0.600	0.392	0.836
3	0.864	0.676	0.352	0.675	0.399	0.839
4	0.764	0.659	0.319	0.667	0.263	0.777
5	0.679	0.661	0.307	0.666	0.418	0.737
6	1.000	0.444	0.267	0.429	0.237	0.641
7	0.963	0.508	0.286	0.495	0.282	0.735
8	0.856	0.632	0.321	0.639	0.358	0.813
9	0.767	0.635	0.302	0.640	0.246	0.754
10	0.683	0.639	0.296	0.638	0.393	0.712
11	0.994	0.190	0.114	0.152	0.060	0.305
12	0.979	0.289	0.165	0.247	0.107	0.462
13	0.861	0.551	0.274	0.537	0.269	0.738
14	0.776	0.589	0.275	0.589	0.223	0.711
15	0.701	0.597	0.275	0.597	0.357	0.673
16	0.993	0.068	0.037	0.103	0.036	0.194
17	0.984	0.176	0.090	0.195	0.075	0.408
18	0.882	0.463	0.220	0.441	0.197	0.668
19	0.781	0.542	0.247	0.534	0.195	0.661



Table 7 (continued)

Simulated dataset	True positive rates					
	8 TPR upper bound (A = 1)	9 Actual TPR(A = 0)	10 Naïve est of TPR(A = 0)	11 Approx TPR(A = 0)	12 TPR lower bound (A = 0)	13 TPR upper bound (A = 0)
20	0.721	0.557	0.252	0.550	0.321	0.628
21	1.000	0.561	0.406	0.504	0.340	0.661
22	0.865	0.581	0.358	0.530	0.334	0.741
23	0.832	0.681	0.359	0.683	0.415	0.837
24	0.755	0.664	0.324	0.677	0.274	0.783
25	0.688	0.672	0.313	0.675	0.426	0.745
26	0.924	0.384	0.260	0.425	0.238	0.587
27	0.907	0.460	0.283	0.447	0.252	0.648
28	0.843	0.629	0.324	0.638	0.361	0.806
29	0.762	0.650	0.314	0.652	0.261	0.763
30	0.691	0.652	0.303	0.655	0.407	0.728
31	0.981	0.183	0.132	0.155	0.064	0.265
32	0.948	0.295	0.181	0.367	0.177	0.553
33	0.877	0.542	0.265	0.531	0.262	0.730
34	0.772	0.597	0.280	0.598	0.229	0.717
35	0.721	0.605	0.279	0.600	0.363	0.677
36	0.991	0.048	0.027	0.049	0.016	0.092
37	0.969	0.153	0.081	0.159	0.059	0.352
38	0.874	0.474	0.227	0.456	0.207	0.676
39	0.786	0.556	0.255	0.541	0.202	0.668
40	0.738	0.580	0.264	0.583	0.347	0.657
41	0.808	0.469	0.339	0.448	0.278	0.598
42	0.836	0.523	0.336	0.495	0.305	0.681
43	0.822	0.672	0.353	0.679	0.407	0.835
44	0.738	0.683	0.337	0.688	0.287	0.792
45	0.706	0.689	0.324	0.692	0.442	0.760
46	0.906	0.340	0.236	0.407	0.220	0.560
47	0.929	0.414	0.247	0.360	0.184	0.576
48	0.824	0.639	0.331	0.627	0.356	0.800
49	0.751	0.663	0.321	0.672	0.270	0.779
50	0.708	0.675	0.315	0.677	0.428	0.747
51	0.958	0.167	0.117	0.147	0.059	0.253
52	0.931	0.287	0.174	0.318	0.147	0.512
53	0.842	0.559	0.279	0.550	0.280	0.745
54	0.779	0.622	0.296	0.628	0.242	0.745
55	0.718	0.647	0.301	0.649	0.401	0.722
56	0.953	0.088	0.059	0.115	0.042	0.181
57	0.942	0.201	0.120	0.206	0.084	0.368
58	0.864	0.510	0.247	0.498	0.237	0.703
59	0.799	0.593	0.276	0.591	0.224	0.713
60	0.744	0.607	0.279	0.603	0.365	0.679
Median absolute error			0.288	0.009		
Median absolute percent error			50.44	2.08		



Table 7 (continued)

Simulated dataset	Equal opportunity					Support info			
	14 Actual <i>EO</i>	15 Naïve est of <i>EO</i>	16 Approx <i>EO</i>	17 <i>EO</i> lower bound	18 <i>EO</i> upper bound	19 Q_{11}	20 Q_{21}	21 Q_{10}	22 Q_{20}
1	0.155	0.038	0.402	0.221	0.591	0.057	– 0.007	0.059	0.014
2	0.047	– 0.011	0.187	– 0.341	0.549	0.061	0.007	0.067	0.018
3	0.015	– 0.035	0.040	– 0.613	0.465	0.219	0.037	0.211	0.044
4	0.003	– 0.030	– 0.024	– 0.399	0.501	0.383	0.091	0.399	0.085
5	– 0.034	– 0.028	– 0.048	– 0.269	0.260	0.579	0.148	0.552	0.122
6	0.408	0.201	0.571	0.359	0.763	0.085	– 0.003	0.065	0.020
7	0.301	0.144	0.369	– 0.144	0.681	0.100	0.010	0.079	0.022
8	0.092	0.015	0.068	– 0.622	0.498	0.238	0.048	0.219	0.047
9	0.050	0.004	0.027	– 0.323	0.521	0.423	0.097	0.405	0.090
10	– 0.005	– 0.012	– 0.012	– 0.222	0.290	0.593	0.156	0.549	0.131
11	0.732	0.431	0.825	0.544	0.934	0.145	0.004	0.052	0.025
12	0.614	0.335	0.688	0.139	0.871	0.169	0.011	0.071	0.029
13	0.223	0.095	0.214	– 0.358	0.592	0.299	0.057	0.216	0.056
14	0.123	0.052	0.116	– 0.188	0.553	0.493	0.107	0.410	0.100
15	0.063	0.023	0.056	– 0.131	0.344	0.632	0.163	0.557	0.141
16	0.884	0.549	0.878	0.274	0.957	0.197	0.005	0.107	0.024
17	0.753	0.445	0.763	0.201	0.909	0.221	0.011	0.097	0.029
18	0.349	0.171	0.355	– 0.110	0.686	0.349	0.062	0.219	0.065
19	0.189	0.096	0.192	– 0.072	0.586	0.548	0.120	0.413	0.109
20	0.132	0.061	0.132	– 0.033	0.400	0.673	0.169	0.562	0.153
21	0.147	0.007	0.496	0.339	0.660	0.041	– 0.001	0.054	0.019
22	0.150	0.025	0.149	– 0.574	0.530	0.053	0.013	0.068	0.021
23	– 0.005	– 0.061	– 0.018	– 0.517	0.417	0.189	0.040	0.211	0.042
24	– 0.004	– 0.046	– 0.017	– 0.299	0.481	0.405	0.088	0.401	0.083
25	– 0.045	– 0.059	– 0.052	– 0.229	0.262	0.558	0.136	0.548	0.121
26	0.454	0.208	0.374	– 0.326	0.686	0.067	0.012	0.070	0.019
27	0.352	0.145	0.313	– 0.534	0.655	0.083	0.017	0.077	0.024
28	0.094	– 0.005	0.058	– 0.403	0.482	0.220	0.047	0.219	0.047
29	0.023	– 0.029	0.027	– 0.243	0.501	0.434	0.091	0.399	0.088
30	– 0.013	– 0.042	– 0.022	– 0.189	0.284	0.583	0.143	0.551	0.126
31	0.733	0.389	0.793	0.433	0.918	0.132	0.007	0.050	0.026
32	0.585	0.294	0.501	– 0.008	0.771	0.145	0.020	0.100	0.025
33	0.235	0.081	0.239	– 0.154	0.614	0.284	0.051	0.221	0.057
34	0.109	0.025	0.108	– 0.129	0.544	0.490	0.104	0.410	0.098
35	0.071	– 0.002	0.075	– 0.077	0.358	0.629	0.143	0.550	0.141
36	0.898	0.536	0.929	0.835	0.975	0.180	0.006	0.075	0.026
37	0.761	0.432	0.765	0.452	0.910	0.195	0.019	0.088	0.030
38	0.335	0.148	0.348	– 0.009	0.668	0.361	0.060	0.220	0.063
39	0.176	0.064	0.190	– 0.029	0.584	0.530	0.110	0.407	0.108
40	0.117	0.024	0.119	– 0.017	0.391	0.669	0.145	0.568	0.144
41	0.318	0.053	0.176	– 0.253	0.530	0.026	0.010	0.059	0.020
42	0.251	0.020	0.135	– 0.284	0.531	0.041	0.014	0.074	0.023
43	0.029	– 0.082	0.014	– 0.283	0.415	0.213	0.041	0.213	0.043
44	– 0.017	– 0.096	– 0.043	– 0.231	0.451	0.387	0.085	0.393	0.082



Table 7 (continued)

Simulated dataset	Equal opportunity					Support info			
	14 Actual <i>EO</i>	15 Naïve est of <i>EO</i>	16 Approx <i>EO</i>	17 <i>EO</i> lower bound	18 <i>EO</i> upper bound	19 Q_{11}	20 Q_{21}	21 Q_{10}	22 Q_{20}
45	− 0.030	− 0.098	− 0.042	− 0.167	0.264	0.545	0.113	0.544	0.116
46	0.527	0.189	0.362	0.027	0.685	0.058	0.014	0.072	0.019
47	0.436	0.139	0.450	0.097	0.744	0.085	0.015	0.070	0.027
48	0.090	− 0.047	0.078	− 0.217	0.468	0.236	0.047	0.211	0.048
49	0.028	− 0.069	0.001	− 0.181	0.482	0.421	0.087	0.401	0.084
50	− 0.002	− 0.083	− 0.020	− 0.141	0.280	0.565	0.118	0.546	0.120
51	0.761	0.361	0.757	0.587	0.899	0.124	0.014	0.053	0.025
52	0.602	0.260	0.543	0.267	0.784	0.155	0.023	0.090	0.027
53	0.218	0.036	0.207	− 0.071	0.562	0.303	0.055	0.218	0.055
54	0.105	− 0.030	0.091	− 0.083	0.537	0.478	0.088	0.405	0.093
55	0.047	− 0.059	0.024	− 0.090	0.317	0.593	0.124	0.551	0.128
56	0.853	0.455	0.800	0.691	0.911	0.186	0.022	0.084	0.025
57	0.715	0.345	0.693	0.484	0.858	0.217	0.026	0.084	0.030
58	0.299	0.086	0.305	0.040	0.627	0.355	0.056	0.223	0.060
59	0.171	0.006	0.163	− 0.004	0.576	0.525	0.088	0.410	0.099
60	0.122	− 0.026	0.103	− 0.005	0.379	0.626	0.123	0.550	0.140
Median absolute error		0.1368	0.013						
Median absolute percent error		69.60	9.00						

estimator, defined in “[Approximation of the ground truth for assessing model fairness](#)” section and Appendix 6, are included for comparison purposes (Table 7 and Figs. 1, 2, 3). In Table 7, Columns 1–3 are the input parameter values, Columns 4–13 are the *TPR* values for true values, the Naïve estimate (see “[Approximation of the ground truth for assessing model fairness](#)” section or Appendix 6) and our proposed approximations (14) along with their associated bounds (19). Columns 14–18 are the *EO* values for true values, the Naïve estimate and our proposed approximations (20) and their associated bounds (21). Columns 19–22 list the values of Q_{1a} and Q_{2a} (from Theorem 1) which are the corresponding group level treatment effects for the non-targeted (bottom 70%) and targeted (top 30%) deciles, respectively.

The results indicate that our proposed approximations for *TPR* and the *EO* metrics are much more accurate than those from the naïve estimator as measured by median absolute errors and median absolute percent errors (see summary results at the bottom of Table 7 and Figs. 1, 2, 3) over the ranges of values for ϕ , μ and θ . The Naïve estimator appears to be biased in that it systematically underestimates both the *TPR* and the *EO* (Figs. 1, 2, 3). As expected, our approximation does not perform as well for 3 out of the 60 data sets (1, 6, and 21) generated with the lowest level of

the treatment effect parameter μ , causing group level treatment effects Q_{21} (Column 20 of Table 7)¹⁵ to become negative which violates Assumption 1.

A case study

To illustrate how the proposed methodology works in the real world, we carry out an end-to-end fairness test using Kevin Hillstrom’s MineThatData E-mail campaign data which has been analyzed by researchers such as Kane et al. (2014), Lo and Pachamanova (2015), and Pachamanova et al. (2020) in another context.

Data

The dataset contains 64,000 customers who made purchases within the past 12 months. The customers were randomly divided into three treatment groups:

- (1) 1/3 to receive email featuring men’s merchandise

¹⁵ When $Q_{21} < 0$ in those special cases, from Lemma 1, it implies that the probability of having a negative lift is greater than that of having a positive lift, i.e., $\delta_{21} > \lambda_{21}$, contrary to general marketing expectations and violating Assumption 1.



Fig. 1 Actual, approximate, and Naïve estimates of true positive rates for $A = 1$, by simulated dataset (x axis) (see Table 7 for the corresponding simulated datasets)

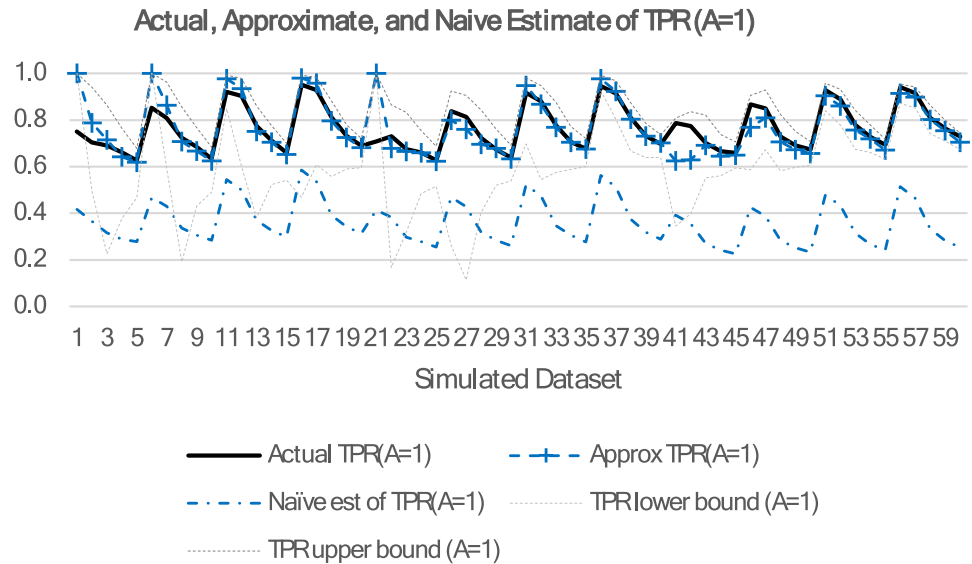
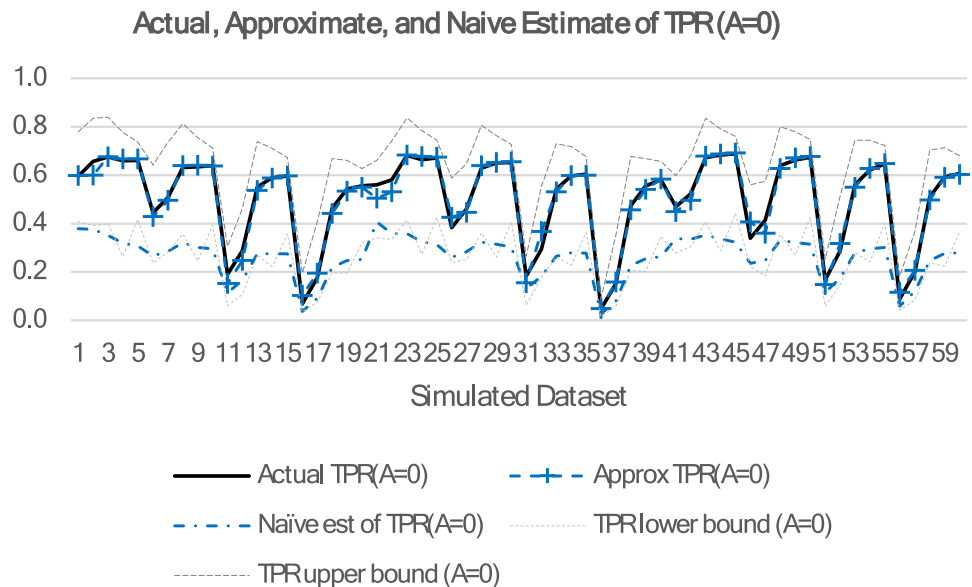


Fig. 2 Actual, approximate, and Naïve estimates of true positive rates for $A = 0$, by case (x axis) (see Table 7 for the corresponding cases)



- (2) 1/3 to receive email featuring women's merchandise
- (3) 1/3 set aside as no email control

Only Groups 2 and 3 are used for subsequent modeling and they are coded as treatment and control, respectively. The target variable is set to 1 if customer visited website within two weeks of email delivery and 0 otherwise. Since the dataset does not include any data element to indicate “protected” groups which is required for fairness testing, for our illustration we create a binary indicator as a proxy¹⁶ for gender based on if a customer purchased Womens merchandise (= 1) in the past year or not (= 0).

¹⁶ This indicator may not be a strong proxy for actual gender and is served as an illustration only.

Uplift modeling

Uplift models based on three alternative methodologies¹⁷ are developed using scikit-uplift Python package from Liksakov (2023):

- Transformation model
- Solo model
- Two-model model

¹⁷ The transformation model is based on a specific transformation on the outcome variable so that regular supervised learning can be utilized (Jaskowski and Jaroszewicz 2012; Kane et al. 2014). The solo model approach uses a treatment dummy as a feature (Lo 2002). The two-model approach is based on two separate models for treatment and control groups as applied in “Empirical evaluation of approaches” section (Lo and Pachamanova 2015; Pachamanova et al. 2020).



The dataset is split into 50% training to build the model and 50% test to report model performance including fairness metrics (as in “[Empirical evaluation of approaches](#)” section). Two types of models are created:

- (A) use all features including the gender proxy
- (B) use all features but omit the gender proxy

Fairness testing

The model performance metric as well as proposed fairness metrics are summarized for Type A models and Type B models in Table 8. Uplift@30 is the observed lift (treatment response rate minus control response rate) in the top 3 deciles for measuring model accuracy.

When gender is used as a model feature (Type A), the approximate EO (i.e., $\widehat{EO}$) values (as well as their bounds) of all three models exceed acceptance range of ± 0.20 . On the other hand, when gender is not included (Type B), the model based on the transformation approach shows a much lower approximate EO value and its bounds are within the acceptance range, leading to a fairness conclusion with this model.¹⁸

Conclusion and discussion

With the rise of ethical AI or responsible AI, quantitative metrics are used with growing popularity to assess fairness for predictive models. Most of these metrics require ground truth of actual individual outcomes which are readily available in standard predictive modeling or response modeling but are unknown in uplift modeling [*Fundamental Problem of Causal Inference* from Holland (1986)]. To our knowledge, no methodology is currently available in the literature for evaluating uplift model fairness without requiring additional data from a subsequent experiment and the construction of surrogate ground truth based on uplift model scores.

In this paper we propose a methodology to derive fairness metrics for uplift models through probability-based approximations. Empirical analysis based on simulation indicates that the approximated metrics are close in value to the true metrics as long as the probability of having a positive lift is greater than the probability of having a

negative lift. Such condition is usually true in marketing and sales applications (e.g., direct marketing, online advertising or personalized recommendations) but may not be valid in some other fields (e.g., medical treatments can have negative effects). Future research is warranted for scenarios where treatments may have strong negative effects such as in the field of personalized medicine. Since treatments are meant to generate a desirable effect by design, we expect our assumption to be valid in most situations. However, if a treatment can truly or hypothetically lead to a strong negative effect with a probability higher than that of having a positive effect (e.g., a bad advertisement, a counterproductive political campaign, or possibly a strong negative effect for a minority subgroup), a different methodological approach will be needed. On the other hand, if the strong negative effect is on another metric (e.g., negative side effects of medicine or medical procedures), a potential research direction is to consider multiple outcome metrics (i.e., an expected positive outcome on the primary metric and a possible negative side effect on another metric) and employ a similar methodology along with multi-objective optimization. Another potential extension is to consider statistical hypothesis testing in conjunction with fairness metrics (e.g., Lo et al. 2024).

The key contribution of our approach is that it will provide a practical methodology for the AI community to evaluate the fairness of uplift models, enabling AI governance (e.g., White House Blueprint for an AI Bill of Rights 2022 and Schwartz et al. 2022) which is rapidly becoming a new standard across industries (e.g., ISO/IEC 2023; Groves et al. 2024; Senate Bill 24-205 2024).

We should caution that the model approval decisions should not be made solely because one or two fairness metrics are out of range (e.g., Meng 2020) due to the fact that it is impossible to simultaneously meet all fairness requirements (e.g., Kleinberg et al. 2017 and Barocas et al. 2023, Chap. 3). An ethics review team with multidisciplinary knowledge and diverse experience is generally helpful in considering fairness requirements and evaluating outcomes (e.g., Sandler and Basl 2017; European Commission 2019; White House Blueprint for an AI Bill of Rights 2022; Blackman 2022, Chap. 6, and Schuett et al. 2024).

Appendix 1: Proof of Lemma 1

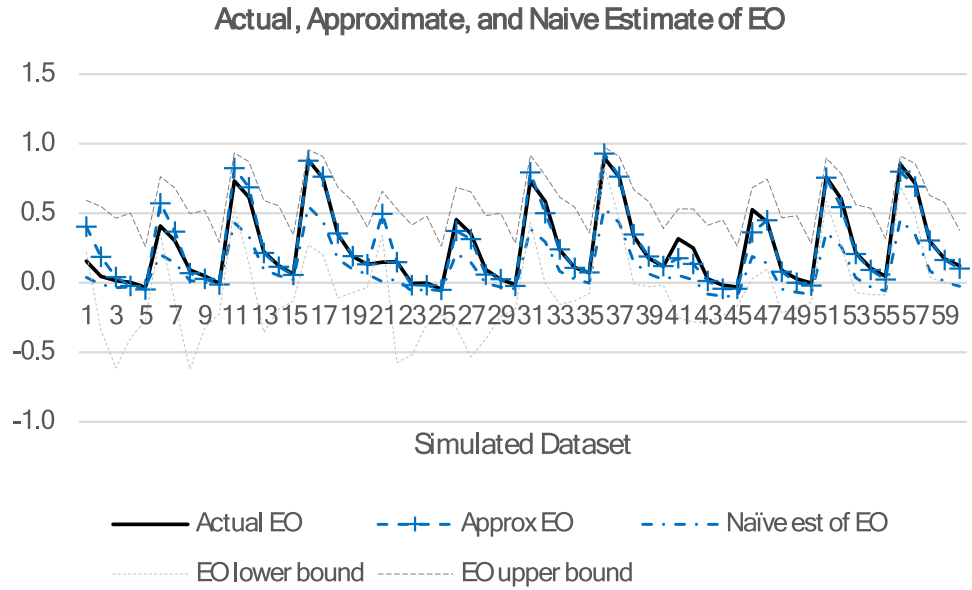
Consider the 2×2 joint probability distribution of the potential outcomes (Y_1, Y_0) in Table 5.

The four quadrants in Table 5 represent the four possible outcomes of (Y_1, Y_0) which are unobservable in practice. Our main interest is in Quadrant C which represents

¹⁸ This example demonstrates that better fairness does not always have to come from a lower accuracy – under the Transformation approach, removing gender from the model results in a much better fairness (in terms of equal opportunity) with a marginal decrease in accuracy (measured by Uplift@30). Additionally, if we compare across the three methods in the lower portion of Table 8, the Transformation approach achieves both the highest accuracy and best fairness. As a result, the relationship between accuracy and fairness is not always negative.



Fig. 3 Actual, approximate, and Naïve estimates of Equal Opportunity, by case (x axis) (see Table 7 for the corresponding cases)



the individual lift, L . Since any real data do not provide *both* potential outcomes, we can only approximate the probability of belonging to Quadrant C using the marginal probabilities of Y_1 and Y_0 , which are estimable. In Table 5, it can be seen that

$$P(Y_1 = 1|H_{ja}) - P(Y_0 = 1|H_{ja})$$

= Probability that an individual belongs to Quadrant C or D
– Probability that an individual belongs to Quadrant B or D.

$$\begin{aligned} &= P(C|H_{ja}) + P(D|H_{ja}) - P(B|H_{ja}) - P(D|H_{ja}) \\ &= P(C|H_{ja}) - P(B|H_{ja}) \\ &= P(Y_1 = 1, Y_0 = 0|H_{ja}) - P(Y_1 = 0, Y_0 = 1|H_{ja}) \\ &= P(L = 1|H_{ja}) - \delta_{ja}, \end{aligned}$$

where δ_{ja} is defined as $P(Y_1 = 0, Y_0 = 1|H_{ja})$, which is Quadrant B in Table 5.

Therefore,

$$\begin{aligned} \lambda_{ja} &= P(L = 1|H_{ja}) \\ &= P(Y_1 = 1|H_{ja}) - P(Y_0 = 1|H_{ja}) + \delta_{ja} \text{ as in (7).} \\ \lambda_{ja} &\geq P(Y_1 = 1|H_{ja}) - P(Y_0 = 1|H_{ja}) \text{ as in (8).} \end{aligned}$$

This bound is also proposed in Tian and Pearl (2000) and Mueller and Pearl (2022).

Appendix 2: Proof of Lemma 2

To support our proof for the bounds, we first consider the following assumptions:

Table 8 Results of Type A models (with gender)

	Transformation	Two model	Solo model
Uplift@30	0.0661	0.0700	0.0661
TPR womens = 1	0.3800	0.2924	0.2880
TPR womens = 0	0.1001	0.0000	0.0514
$\widehat{EO}$	0.2799	0.2924	0.2365
Upper EO Bound	0.3250	0.3302	0.2828
Lower EO Bound	0.2330	0.2680	0.2017
	Transformation	Two model	Solo model
Uplift@30	0.0653	0.0616	0.0601
TPR womens = 1	0.2859	0.2951	0.2793
TPR womens = 0	0.2343	0.0050	0.0120
$\widehat{EO}$	0.0516	0.2900	0.2673
Upper EO bound	0.1059	0.3277	0.3035
Lower EO bound	0.0012	0.2638	0.2396

Results of Type B models (without gender)

Assumption A2.1¹⁹

$$P(Y_1 = 1|Y_0 = 1, H_{ja}) \geq P(Y_1 = 0|Y_0 = 0, H_{ja}).$$

This assumption means that if an untreated person responds, the probability of their response IF treated is greater than or equal to the opposite response when the

¹⁹ We consider Assumptions A2.1 and A2.2 reasonable and they are much weaker than the monotone treatment response assumption by Manki (1997), where $Y_1 \geq Y_0$ for *every* individual (see Flores and Chen, 2018, Chap. 3) and Neal (2020, Chap. 8) as related literature).



Table 9 Observed take-action (Y) vs true ITE (L)

Actual outcome	Hypothetical outcome (what– if)	Observed (Y) vs true (L)	Naïve estimator result
$Y = 1$ if treated	$Y = 0$ if not treated	$Y = 1$ and $L = 1$	Correct
	$Y = 1$ if not treated	$Y = 1$ and $L < 1$	Incorrect
$Y = 0$ if treated	$Y = 0$ if not treated	$Y = 0$ and $L < 1$	Correct
	$Y = 1$ if not treated	$Y = 0$ and $L < 1$	Correct
$Y = 1$ if not treated	$Y = 0$ if treated	$Y = 1$ and $L < 1$	Incorrect
	$Y = 1$ if treated	$Y = 1$ and $L < 1$	Incorrect
$Y = 0$ if not treated	$Y = 0$ if treated	$Y = 0$ and $L < 1$	Correct
	$Y = 1$ if treated	$Y = 0$ and $L = 1$	Incorrect

untreated person does not respond. An interpretation is that if an untreated person does not respond ($Y_0 = 0$), it is uncertain if they would respond or not if treated (i.e., Y_1 can be 0 or 1). But if the untreated person does respond ($Y_0 = 1$), it is more likely the same person would respond ($Y_1 = 1$) if treated as we expect the treatment to have a positive impact. Likewise, we have a similar but parallel assumption below given the knowledge of Y_1 .

Assumption

A2.2 $P(Y_0 = 0|Y_1 = 0, H_{ja}) \geq P(Y_0 = 1|Y_1 = 1, H_{ja})$.

This means that if a treated person does not respond ($Y_1 = 0$), it is most likely that the same person would also not respond if untreated ($Y_0 = 0$). If the treated person responds instead ($Y_1 = 1$), the same person may or may not respond if untreated (Y_0 can be 0 or 1).

We can now proceed with the proof for Lemma 2 .

$$\begin{aligned}
 \frac{\lambda_{ja}}{\delta_{ja}} &= \frac{P(Y_1 = 1, Y_0 = 0|H_{ja})}{P(Y_1 = 0, Y_0 = 1|H_{ja})} \\
 &= \frac{P(Y_0 = 0|H_{ja})P(Y_1 = 1|Y_0 = 0, H_{ja})}{P(Y_0 = 1|H_{ja})P(Y_1 = 0|Y_0 = 1, H_{ja})} \\
 &= \left(\frac{1 - p_{cja}}{p_{cja}} \right) \frac{1 - P(Y_1 = 0|Y_0 = 0, H_{ja})}{1 - P(Y_1 = 1|Y_0 = 1, H_{ja})} \\
 &\geq \frac{1 - p_{cja}}{p_{cja}} \text{ by Assumption A2.1,}
 \end{aligned} \tag{25}$$

where $p_{cja} = P(Y_0 = 1|H_{ja})$.

Similarly,

$$\begin{aligned}
 \frac{\lambda_{ja}}{\delta_{ja}} &= \frac{P(Y_1 = 1, Y_0 = 0|H_{ja})}{P(Y_1 = 0, Y_0 = 1|H_{ja})} \\
 &= \frac{P(Y_1 = 1|H_{ja})P(Y_0 = 0|Y_1 = 1, H_{ja})}{P(Y_1 = 0|H_{ja})P(Y_0 = 1|Y_1 = 0, H_{ja})} \\
 &= \left(\frac{p_{tja}}{1 - p_{tja}} \right) \frac{1 - P(Y_0 = 1|Y_1 = 1, H_{ja})}{1 - P(Y_0 = 0|Y_1 = 0, H_{ja})} \\
 &\geq \frac{p_{tja}}{1 - p_{tja}} \text{ by Assumption A2.2,}
 \end{aligned} \tag{26}$$

where $p_{tja} = P(Y_1 = 1|H_{ja})$.

Combining (25) and (26), we have:

$$\frac{\lambda_{ja}}{\delta_{ja}} \geq \text{Max} \left(\frac{p_{tja}}{1 - p_{tja}}, \frac{1 - p_{cja}}{p_{cja}} \right)$$

as in (15).

Appendix 3: Proof of Theorem 2

Recall from Eq. (12):

from Lemma 1

$$\begin{aligned}
 TPR(A=a) &= P(\text{Top } \alpha \text{ ranked by } S|L=1, A=a) = P(S \geq k|L=1, A=a) \\
 &= \frac{P(L=1|S \geq k, A=a)P(S \geq k|A=a)}{P(L=1|S \geq k, A=a)P(S \geq k|A=a) + P(L=1|S < k, A=a)P(S < k|A=a)} \\
 &= \frac{[P(Y_1 = 1|H_{1a}) - P(Y_0 = 1|H_{1a}) + \delta_{1a}]P(S \geq k|A=a)}{[P(Y_1 = 1|H_{1a}) - P(Y_0 = 1|H_{1a}) + \delta_{1a}]P(S \geq k|A=a) + [P(Y_1 = 1|H_{2a}) - P(Y_0 = 1|H_{2a}) + \delta_{2a}]P(S < k|A=a)}
 \end{aligned}$$



$$= \frac{1}{1 + \frac{Q_{2a} + \delta_{2a}}{Q_{1a} + \delta_{1a}} O_{Sa}} \quad (12)$$

where H_{1a} = the set of $\{S \geq k, A = a\}$, and $\delta_{1a} = P(Y_1 = 0, Y_0 = 1 | H_{1a}) \geq 0$, H_{2a} = the set of $\{S < k, A = a\}$, and $\delta_{2a} = P(Y_1 = 0, Y_0 = 1 | H_{2a}) \geq 0$, $Q_{2a} = P(Y_1 = 1 | H_{2a}) - P(Y_0 = 1 | H_{2a})$, $Q_{1a} = P(Y_1 = 1 | H_{1a}) - P(Y_0 = 1 | H_{1a})$, and $O_{Sa} = \frac{1 - P(S \geq k | A = a)}{P(S \geq k | A = a)}$.

Note that Q_{1a} and Q_{2a} only involve marginal probabilities and, thus, are estimable from data, but δ_{1a} and δ_{2a} are not because they involve bivariate probabilities of the potential outcomes.

Recall:

$$\begin{aligned} \lambda_{2a} &= P(Y_1 = 1, Y_0 = 0 | H_{2a}), \lambda_{1a} \\ &= P(Y_1 = 1, Y_0 = 0 | H_{1a}). \end{aligned}$$

Consider the relative difference between Eqs. (12) and (13):

$$\begin{aligned} \frac{TPR(A = a) - \widehat{TPR}(A = a)}{TPR(A = a)} &= \frac{\frac{Q_{2a}}{Q_{1a}} O_{Sa} - \frac{Q_{2a} + \delta_{2a}}{Q_{1a} + \delta_{1a}} O_{Sa}}{1 + \frac{Q_{2a}}{Q_{1a}} O_{Sa}} \\ &= \kappa_a \left(1 - \frac{1 + \delta_{2a}/Q_{2a}}{1 + \delta_{1a}/Q_{1a}} \right) \\ &= \kappa_a \left(\frac{\frac{1}{\delta_{1a}} - 1 - \frac{1}{\delta_{2a}} - 1}{1 + \frac{1}{\delta_{1a}} - 1} \right) \\ &= \kappa_a \left(\frac{1 - \frac{\delta_{1a}}{\delta_{2a}}}{\frac{\delta_{1a}}{\delta_{2a}} - 1} \right) \end{aligned}$$

$\leq \frac{\kappa_a}{\frac{\delta_{1a}}{\delta_{2a}}}$, since $\lambda_{ja} > \delta_{ja}$ is implied by Lemma 2 when $p_{tja} > p_{cja}, j = 1, 2$ (also stated in Assumption 1 below Lemma 1).

$\leq \frac{\kappa_a}{M_{1a}}$, applying Lemma 2, which is the upper bound in (17), where $M_{ja} = \max\left(\frac{p_{tja}}{1 - p_{tja}}, \frac{1 - p_{cja}}{p_{cja}}\right)$ and $\kappa_a =$

$$\frac{1}{(1 + \frac{Q_{2a}}{Q_{1a}} O_{Sa})} \frac{Q_{2a}}{Q_{1a}} O_{Sa} \text{ or}$$

for practical computation.

Rearranging the latter and substituting from (13), we have the upper bound in (16):

$$TPR(A = a) \leq \frac{\widehat{TPR}(A = a)}{1 - \frac{\kappa_a}{M_{1a}}} = \frac{1}{1 + \frac{Q_{2a}}{Q_{1a}} \left(1 - \frac{1}{M_{1a}}\right) O_{Sa}}.$$

To prove the lower bound in (16), consider again the relative error,

$$\begin{aligned} \frac{TPR(A = a) - \widehat{TPR}(A = a)}{\widehat{TPR}(A = a)} &= \kappa_a \left(\frac{\frac{\delta_{1a}}{Q_{1a}} - \frac{\delta_{2a}}{Q_{2a}}}{1 + \frac{\delta_{1a}}{Q_{1a}}} \right) = \\ \kappa_a \left(\frac{\frac{\delta_{1a} - \frac{Q_{1a}\delta_{2a}}{Q_{2a}}}{Q_{1a} + \delta_{1a}}}{1 + \frac{\delta_{1a}}{Q_{1a}}} \right) &\cong \kappa_a \left(\frac{\delta_{1a}}{Q_{1a}} - \frac{\delta_{2a}}{Q_{2a}} \right), \text{ using the two-dimensional} \\ \text{first order Taylor series expansion about } (\delta_{1a} = 0, \delta_{2a} = 0) \\ \text{on } \frac{\delta_{1a} - \frac{Q_{1a}\delta_{2a}}{Q_{2a}}}{Q_{1a} + \delta_{1a}}, \frac{TPR(A = a) - \widehat{TPR}(A = a)}{\widehat{TPR}(A = a)} &\geq -\kappa_a \frac{\delta_{2a}}{Q_{2a}} = -\kappa_a \frac{\delta_{2a}}{\lambda_{2a} - \delta_{2a}} = \\ -\frac{\kappa_a}{\frac{\delta_{2a}}{\lambda_{2a}} - 1} &\geq -\frac{\kappa_a}{M_{2a} - 1}, \text{ by applying Lemma 2, which is the} \\ \text{lower bound in (17) and completes (17).} \end{aligned}$$

Rearranging the latter,

$$TPR(A = a) \geq \frac{\widehat{TPR}(A = a)}{1 + \frac{\kappa_a}{M_{2a} - 1}} = \frac{1}{1 + \frac{Q_{2a}}{Q_{1a}} \left(1 + \frac{1}{M_{2a} - 1}\right) O_{Sa}}.$$

This completes (16).

Appendix 4: Approximation formula and bound for the false positive rate and predictive equality

Similar to the approximation for the True Positive Rate (TPR), we can compute the approximation for the False Positive Rate $FPR(A = a)$ as follows.

$$\begin{aligned} FPR(A = a) &= P(\text{Top } \alpha \text{ ranked by } S | L < 1, A = a) = P(S \geq k | L < 1, A = a) \\ &= \frac{P(L < 1 | S \geq k, A = a) P(S \geq k | A = a)}{P(L < 1 | S \geq k, A = a) P(S \geq k | A = a) + P(L < 1 | S < k, A = a) P(S < k | A = a)} \\ &= \frac{(1 - P(L = 1 | S \geq k, A = a)) P(S \geq k | A = a)}{(1 - P(L = 1 | S \geq k, A = a)) P(S \geq k | A = a) + (1 - P(L = 1 | S < k, A = a)) P(S < k | A = a)} \\ &= \frac{(1 - \lambda_{1a}) P(S \geq k | A = a)}{(1 - \lambda_{1a}) P(S \geq k | A = a) + (1 - \lambda_{2a}) P(S < k | A = a)} = \frac{1}{1 + \frac{1 - \lambda_{2a}}{1 - \lambda_{1a}} O_{Sa}} = \frac{1}{1 + \frac{1 - (Q_{2a} + \delta_{2a})}{1 - (Q_{1a} + \delta_{1a})} O_{Sa}} \quad (27) \end{aligned}$$

$$= \frac{\max\{0, Q_{2a}\} (1 - P(S \geq k | A = a))}{\max\{0, Q_{2a}\} (1 - P(S \geq k | A = a)) + \max\{0, Q_{1a}\} P(S \geq k | A = a)}$$

by (7), where H_{1a} = the set of $\{S \geq k, A = a\}$, and $\delta_{1a} = P(Y_1 = 0, Y_0 = 1 | H_{1a}) \geq 0$, H_{2a} = the set of $\{S < k, A = a\}$, and



$\delta_{2a} = P(Y_1 = 0, Y_0 = 1 | H_{2a}) \geq 0$, $Q_{ja} = P(Y_1 = 1 | H_{ja}) - P(Y_0 = 1 | H_{ja})$ and $O_{Sa} = \frac{1 - P(S \geq k | A = a)}{P(S \geq k | A = a)}$, as before.

We may notice the similarity between (27) and (12). Applying the same approximation from (13) to (27), we have

$$\widehat{FPR}(A = a) = \frac{1}{1 + \frac{1 - Q_{2a}}{1 - Q_{1a}} O_{Sa}} \quad (28)$$

$$\begin{aligned} \frac{1 - Q_{2a}}{1 - Q_{1a}} \text{Max}\{0, \left(1 - \frac{\delta_{2a}}{1 - Q_{2a}}\right)\} &\leq K(\lambda_{1a}, \lambda_{2a}) \\ &= \frac{1 - (Q_{2a} + \delta_{2a})}{1 - (Q_{1a} + \delta_{1a})} \leq \frac{1 - Q_{2a}}{1 - Q_{1a}} \left(1 + \frac{\delta_{1a}}{1 - Q_{1a}}\right) \end{aligned} \quad (32)$$

by setting $\delta_{1a} = 0$ and $\delta_{2a} = 0$, for the lower bound and upper bound, respectively.

Applying (32) to (27) and taking the computational consideration into account similar to (29), we have:

$$\text{or} = \frac{(1 - \max\{0, Q_{1a}\})P(S \geq k | A = a)}{(1 - \max\{0, Q_{1a}\})P(S \geq k | A = a) + (1 - \max\{0, Q_{2a}\})(1 - P(S \geq k | A = a))}, \quad (29)$$

where (29) is a more practical computational formula in situations where the denominator of O_{Sa} is very small, or the estimate for Q_{1a} or Q_{2a} is negative.

Using (29), the *FPR* parity or Predictive Equality (*PE*) can be estimated as follows:

$$\begin{aligned} \widehat{PE} &= \widehat{FPR}(A = 1) - \widehat{FPR}(A = 0) \\ &= \frac{(1 - \max\{0, Q_{11}\})P(S \geq k | A = 1)}{(1 - \max\{0, Q_{11}\})P(S \geq k | A = 1) + (1 - \max\{0, Q_{21}\})(1 - P(S \geq k | A = 1))} \\ &\quad - \frac{(1 - \max\{0, Q_{10}\})P(S \geq k | A = 0)}{(1 - \max\{0, Q_{10}\})P(S \geq k | A = 0) + (1 - \max\{0, Q_{20}\})(1 - P(S \geq k | A = 0))}. \end{aligned} \quad (30)$$

Next, we construct a bound for *PE* similar to (19) for *EO*, consider the following function:

$$K(\lambda_{1a}, \lambda_{2a}) = \frac{1 - \lambda_{2a}}{1 - \lambda_{1a}} = \frac{1 - (Q_{2a} + \delta_{2a})}{1 - (Q_{1a} + \delta_{1a})} \geq 0. \quad (31)$$

Using the two-dimensional first order Taylor series expansion of $K(\lambda_{1a}, \lambda_{2a})$ about (Q_{1a}, Q_{2a}) in (31):

$$\begin{aligned} K(\lambda_{1a}, \lambda_{2a}) &\cong K(Q_{1a}, Q_{2a}) + \frac{\partial K}{\partial \lambda_{1a}}|_{(Q_{1a}, Q_{2a})} \delta_{1a} \\ &\quad + \frac{\partial K}{\partial \lambda_{2a}}|_{(Q_{1a}, Q_{2a})} \delta_{2a} \\ &= K(Q_{1a}, Q_{2a}) + \frac{1 - Q_{2a}}{(1 - Q_{1a})^2} \delta_{1a} - \frac{1}{1 - Q_{1a}} \delta_{2a} \\ &= \frac{1 - Q_{2a}}{1 - Q_{1a}} \left(1 + \left(\frac{\delta_{1a}}{1 - Q_{1a}} - \frac{\delta_{2a}}{1 - Q_{2a}}\right)\right). \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{(1 - \max\{0, Q_{1a}\})P(S \geq k | A = a)}{(1 - \max\{0, Q_{1a}\})P(S \geq k | A = a) + (1 - \max\{0, Q_{2a}\})(1 - P(S \geq k | A = a))} \left(1 + \frac{1}{\frac{1 - Q_{1a}}{1 - Q_{2a}} + 1}\right) &\leq \widehat{FPR}(A = a) \leq \\ \frac{(1 - \max\{0, Q_{1a}\})P(S \geq k | A = a)}{(1 - \max\{0, Q_{1a}\})P(S \geq k | A = a) + (1 - \max\{0, Q_{2a}\})(1 - P(S \geq k | A = a)) \text{Max}\left\{0, \left(1 - \frac{1}{\frac{1 - Q_{1a}}{1 - Q_{2a}} + 1}\right)\right\}} & \end{aligned} \quad (33)$$

In (33),

$$\frac{1 - \lambda_{ja}}{\delta_{ja}} = \frac{1 - (\lambda_{ja} + \delta_{ja})}{\delta_{ja}} + 1 \geq \frac{\lambda_{ja} + \delta_{ja}}{\delta_{ja}} + 1,$$

because $\lambda_{ja} + \delta_{ja} \leq \lambda_{ja} + P(Y_1 = 1, Y_0 = 1 | H_{ja}) = p_{tja} \leq \frac{1}{2}$,

if $P(Y_1 = 1 | Y_0 = 1, H_{ja})$ is assumed to be at least $P(Y_1 = 0 | Y_0 = 1, H_{ja})$, which means that if an untreated individual responds ($Y_0 = 1$), the probability of response ($Y_1 = 1$) if the same person is treated would be higher than that of no response ($Y_1 = 0$) as we expect the treatment to have a positive impact; and p_{tja} is assumed to be at most $\frac{1}{2}$ which generally holds in marketing problems.



Then, using Lemma 2,

$$\frac{1 - \lambda_{ja}}{\delta_{ja}} \geq \frac{\lambda_{ja}}{\delta_{ja}} + 2 \geq M_{ja} + 2 = \text{Max} \left(\frac{p_{tja}}{1 - p_{tja}}, \frac{1 - p_{cja}}{p_{cja}} \right) + 2. \quad (35)$$

Applying (35) to (33), we have

$0.15, M_{1a} = 4; Q_{2a} = 0.05, M_{2a} = 9;$ and $P(S \geq k|A = a) = 0.3$, and using the approximation for FPR in (29),

$$\widehat{FPR}(A = a) = \frac{(1 - 0.15) \times 0.3}{(1 - 0.15) \times 0.3 + (1 - 0.05) \times 0.7} = 0.277.$$

$$\begin{aligned} & \frac{(1 - \max\{0, Q_{1a}\})P(S \geq k|A = a)}{(1 - \max\{0, Q_{1a}\})P(S \geq k|A = a) + (1 - \max\{0, Q_{2a}\})(1 - P(S \geq k|A = a)) \left(1 + \frac{1}{M_{1a}+3}\right)} \\ & \leq FPR(A = a) \leq \frac{(1 - \max\{0, Q_{1a}\})P(S \geq k|A = a)}{(1 - \max\{0, Q_{1a}\})P(S \geq k|A = a) + (1 - \max\{0, Q_{2a}\})(1 - P(S \geq k|A = a)) \text{Max} \left\{0, \left(1 - \frac{1}{M_{2a}+3}\right)\right\}}, \end{aligned} \quad (36)$$

where, from (18), $M_{ja} = \text{Max} \left(\frac{p_{tja}}{1 - p_{tja}}, \frac{1 - p_{cja}}{p_{cja}} \right), j = 1, 2$, which can be simplified as $\frac{1 - p_{cja}}{p_{cja}}$ if $p_{tja} \leq 0.5$ and $p_{cja} \leq 0.5$.

Inequality (36) can be re-expressed as a bound for the relative error for $\widehat{FPR}(A = a)$. Define the upper and lower bound values in (36) as U_{FPR_a} and L_{FPR_a} , respectively. Then rearranging (36),

$$1 - \frac{\widehat{FPR}(A=a)}{L_{FPR_a}} \leq \frac{TPR(A=a) - \widehat{TPR}(A=a)}{TPR(A=a)} \leq 1 - \frac{\widehat{FPR}(A=a)}{U_{FPR_a}}. \quad (37)$$

Following the numerical example near the end of “Approximation of the ground truth for assessing model fairness” section (below Theorem 2), where $Q_{1a} =$

And employing (36), we obtain

$$0.251 \leq FPR(A = a) \leq 0.295.$$

Using (37), we can compute a bound for the relative error for the approximate FPR : $-0.104 \leq \text{relative error} \leq 0.061$.

Finally, to obtain a bound for Predictive Equality (PE), we utilize the left and right limits from (36) for $FPR(A = 1)$ and $FPR(A = 0)$:

$$\begin{aligned} & \frac{(1 - \max\{0, Q_{11}\})P(S \geq k|A = 1)}{(1 - \max\{0, Q_{11}\})P(S \geq k|A = 1) + (1 - \max\{0, Q_{21}\})(1 - P(S \geq k|A = 1)) \left(1 + \frac{1}{M_{11}+3}\right)} \\ & - \frac{(1 - \max\{0, Q_{10}\})P(S \geq k|A = 0)}{(1 - \max\{0, Q_{10}\})P(S \geq k|A = 0) + (1 - \max\{0, Q_{20}\})(1 - P(S \geq k|A = 0)) \text{Max} \left\{0, \left(1 - \frac{1}{M_{20}+3}\right)\right\}} \\ & \leq PE \leq \frac{(1 - \max\{0, Q_{11}\})P(S \geq k|A = 1)}{(1 - \max\{0, Q_{11}\})P(S \geq k|A = 1) + (1 - \max\{0, Q_{21}\})(1 - P(S \geq k|A = 1)) \text{Max} \left\{0, \left(1 - \frac{1}{M_{21}+3}\right)\right\}} \\ & - \frac{(1 - \max\{0, Q_{10}\})P(S \geq k|A = 0)}{(1 - \max\{0, Q_{10}\})P(S \geq k|A = 0) + (1 - \max\{0, Q_{20}\})(1 - P(S \geq k|A = 0)) \left(1 + \frac{1}{M_{10}+3}\right)}. \end{aligned} \quad (38)$$



Appendix 5: Simulation details

4. Age is the individual's age, which follows a normal distribution of mean of 55 and standard deviation of 10, i.e., $Age = 55 + 10 \times \xi$, where $\xi \sim N(0,1)$.
5. Income is related with age exponentially: $Income = \exp(4 + 0.007age + 0.5 \times \xi)$, where $\xi \sim N(0,1)$.
6. Spent is the individual previous spending on the donation, which depends on the income: $Spent = 5 * (income \times 8 + 150 \times \xi)$, where $\xi \sim N(0,1)$.
7. Frequency is the number of donations made in the past by an individual, which follows a binomial distribution: $Frequency \sim Binominal(10,0.25)$.
8. Recency is the individual's recent donation: $Recency \sim Poisson(1)$.
9. Gender is the attribute for the fairness assessment, known as A in the paper: $Gender \sim Bernoulli(0.5)$.

Appendix 6: Description of the naïve estimator

Repeating the definition outlined in “[Approximation of the ground truth for assessing model fairness](#)” section, EO is the difference in True Positive Rate (TPR) between two groups:

$$EO = TPR(A = 1) - TPR(A = 0),$$

where A is a protected attribute such as gender or race, and

$$TPR(A = a) = P(\text{selected targets} \mid \text{actual good targets}, A = a).$$

Assuming a threshold k is established by marketers for selecting a target audience (model score $S \geq k$), then (suppressing subscript i for brevity):

$$TPR(A = a) = P(\text{Top scores ranked by } S \mid L = 1, A = a) = P(S \geq k \mid L = 1, A = a).$$

Given that the true ITE (L) is unobservable, a Naïve estimator can be constructed by using the observed value of Y to represent L as follows:

$$\begin{aligned} \text{Naive estimator of } TPR(A = a) &= \\ P(\text{Top scores ranked by } & \\ S \mid Y = 1, A = a) &= P(S \geq k \mid Y = 1, A = a). \end{aligned}$$

We argue that this estimator is systematically biased because the observed value of Y does not always accurately represent L . Table 9 outlines the eight possible outcomes between the observed Y and the hypothetical,

counterfactual Y . For example, for a person who was treated and he/she took action (i.e., responded), Y is correct to represent L if he/she would not take action if not treated, but it is incorrect if he/she would also take action if not treated. Other scenarios can be interpreted similarly. It turns out that Y only represents L accurately in 4 out of the 8 possible outcomes.

Data availability Data available upon request.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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